Biology 1002 R Assignment
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Moose Study:

```
install.packages("dplyr")
> library("dplyr")
library(dplyr)
> > install.packages(dplyr)
> library(readr)
> MoosePopulation <- read_csv("Moose/MoosePopulation.csv")
> View(MoosePopulation)
> MoosePopulation <- na.omit(MoosePopulation)
> MoosePopulation <- select(MoosePopulation, Ecoregion, Year, Area,
Estimated_Moose_Pop)
> oldest_year<-min(MoosePopulation$Year)
> print(oldest_year)
[1] 1904
> highest_moose_population<-max(MoosePopulation$Estimated_Moose_Pop)
> print(highest_moose_population)
[1] 41250
> #Question5a_1904
> #Question5B_41250
> > MoosePopulation <- mutate(MoosePopulation, MooseDensity =
Estimated_Moose_Pop/Area)
> plot(MoosePopulation$Year, MoosePopulation$MooseDensity,
+    xlab = "year",
+    ylab = "Moose per sq km",
+    main = "Moose density in Newfoundland ecoregions over time")
> MoosePopulationWest <- filter(MoosePopulation, Ecoregion == "Western_Forests")
> plot(MoosePopulationWest$Year, MoosePopulationWest$MooseDensity,
+    xlab = "year",
+    ylab = "Moose per sq km",
+    main = "Moose density in Newfoundland ecoregions over time","l")
> MoosePopulation2020 <- arrange(MoosePopulation, desc(MooseDensity))
> print(MoosePopulation2020)
> moosefinal <- MoosePopulation2020 %>%
+   filter(Year == 2020) %>%
```

```
+   filter(MooseDensity > 2.0) %>%
+   arrange(desc(MooseDensity)) %>%
+   print(.)
```

Sapling Study

```
Q11. > library(readr)
> SaplingStudy <- read_csv("Moose/SaplingStudy.csv")

> View(SaplingStudy)
> saplings <- SaplingStudy
> sap_clean <- na.omit(saplings)

library(dplyr)
Q12. > sap_reg_browse <- sap_clean %>%
+ group_by(Ecoregion) %>%
+ summarize(AverageBrowsing = mean(BrowsingScore, na.rm = TRUE))

> print(sap_reg_browse)

> avg_browse_reg <- sap_reg_browse %>%
+ arrange(desc(AverageBrowsing))
> print(avg_browse_reg)

> #Northern_Peninsula_Forests_Highest_4.57
> #StraitOfBelleIsleBarrens_Lowest_1

Q13. > sap_reg_height_low <- saplings %>%
+ group_by(Ecoregion) %>%
+ summarise(Height = mean(Height, na.rm = TRUE)) %>%
+ filter(Height < 20)
> print(sap_reg_height_low)

#Northern_Peninsula_Forests19.9cm

#Western_Forests18.9cm

Q14. > sap_spe_browse <- saplings %>%
+ group_by(Species) %>%
+ summarise(BrowsingScore = mean(BrowsingScore, na.rm = TRUE))
> print(sap_spe_browse)
```

```
 > avg_browse_spe <- sap_spe_browse %>%
+ arrange(desc(BrowsingScore))
> print(avg_browse_spe)
```

#Black_Ash_BrowsingScore_5

#Black_Spruce_BrowsingScore_2.33

Q15.
```
> fir_reg_browse <- saplings %>%
+ group_by(Ecoregion) %>%
+ filter(Species == "Balsam_Fir") %>%
+ summarise(BrowsingScore = mean(BrowsingScore, na.rm = TRUE))
> print(fir_reg_browse)
```

Q16.
```
> barplot(fir_reg_browse$BrowsingScore, names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion", ylab = "Average_Balsam_Browsing_Score", main =
"Average_Balsam_Fir_Browsing_by_Ecoregion", col = "brown", cex.names = 0.6)
```

Q17.
```
> spruce_reg_browse <- saplings %>%
+ filter(Species == "Black_Spruce") %>%
+ group_by(Ecoregion) %>%
+ summarise(Browsing = mean(BrowsingScore, na.rm = TRUE))
> spruce_reg_browse$Browsing <- as.numeric(spruce_reg_browse$Browsing)
> print(spruce_reg_browse)
```

```
> barplot(spruce_reg_browse$Browsing, names.arg = spruce_reg_browse$Ecoregion, xlab
= "Ecoregion", ylab = "Average Black Spruce Browsing Score", main = "Average Black
Spruce Browsing by Ecoregion", col = "lightpink", cex.names = 0.6)
```

#BrowsingInBlackSpruceIsLower_MoosePreferBalsamFir

Q18.
```
sap_reg_tally<- sap_clean %>%

 group_by(Ecoregion) %>%

 tally() %>%

 print()
> #There is a different # of trees counted in each ecoregion ranging from 1-9
```

Q19.
```
> sap_spe_tally<- sap_clean %>%
+   group_by(Species) %>%
+   tally() %>%
+   print()
```

> #For each species it ranged from 1-11, BalsamFir being the highest and BlackAsh the lowest

Q20.

A) #I think that the Sapling Study is unevenly distributed based on what ecoregion and sapling species are available at the given moment. Some ecoregions are more commonly found than others. Similarly, some sapling species are more commonly found. These are overrepresented in the study while rarer species and ecoregions are underrepresented.

B) #It is important to recognize bias in ecological datasets because individuals and scientists can come to the wrong conclusion about species and their specific interactions in the world. Recognizing bias in ecological datasets supports future ecological decisions and shows true care for the ecosystem being studied.

Q21a)

```
> moose_2020b <- MoosePopulation %>%
+ filter(Year == 2020) %>%
+ mutate(MooseDensity = Year/Area)
> print(moose_2020b)
```

Q21b) >moose_sap <- left_join(moose_2020b, sap_clean, by = "Ecoregion", relationship = "many-to-many")

```
> print(moose_sap)
```

Q22) > sum_spe_browse <- moose_sap %>%

```
+ group_by(Species, Ecoregion) %>%
+ summarize(mean_BrowsingScore = mean(BrowsingScore),mean_MooseDensity = mean(MooseDensity))
> print(sum_spe_browse)
```

Q23) > sum_spe_browse <- moose_sap %>%

```
+ group_by(Species, Ecoregion) %>%
+ summarize(mean_BrowsingScore = mean(BrowsingScore),mean_MooseDensity = mean(MooseDensity))
> print(sum_spe_browse)
```

a) The graph does support the hypothesis of the researchers. When the moose density is low, most moose browsing is strongly preferred for the willow species. When the moose density is high, the moose browsing shows a spread across species, supporting generalist browsing.

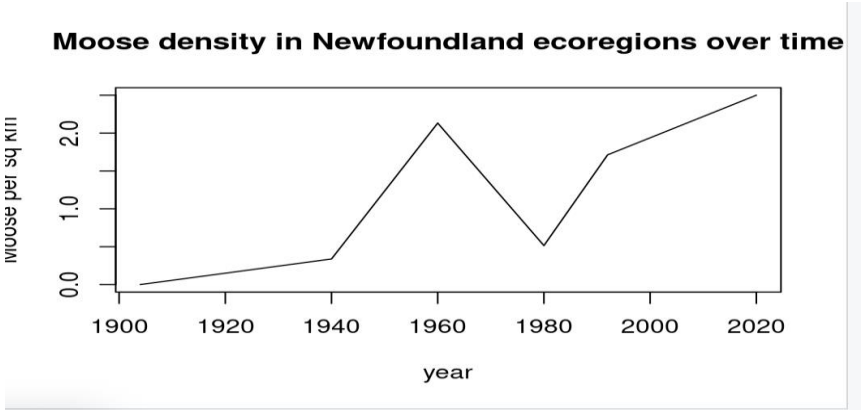
b) The moose favors the willow species the most since it has the highest average browsing scores. The moose browses the black spruce the least since this species has the lowest average browsing scores.

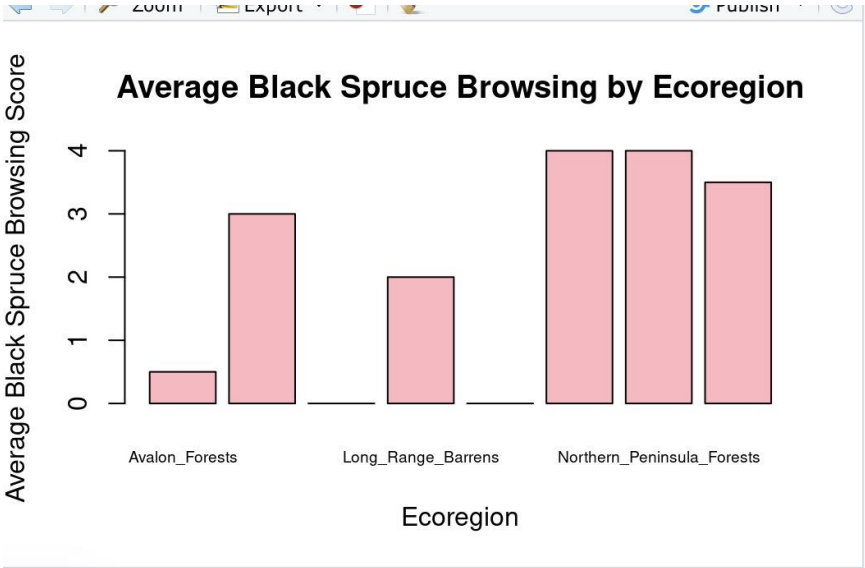
c) The species missing from the graph is black ash, probably because there was not enough data for the graph to interpret.

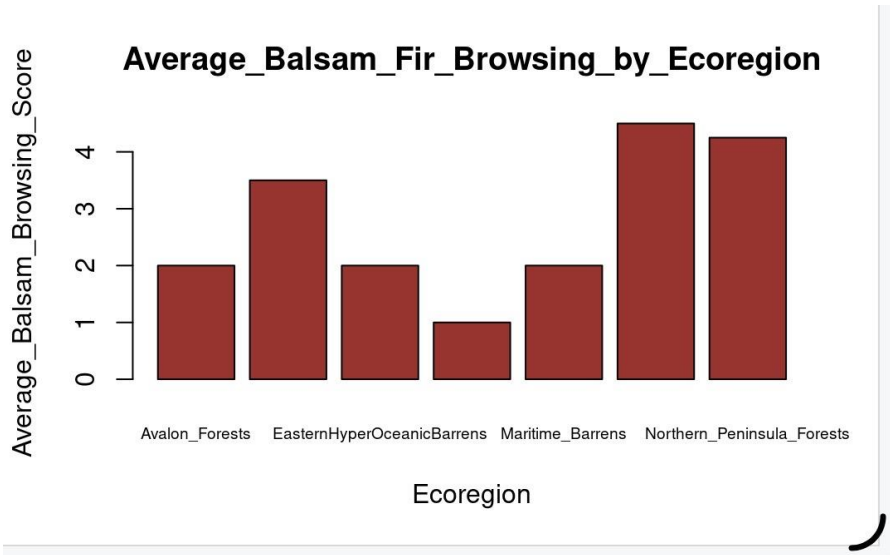
Q24) collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
human_pop <- c(18000, 12000, 4000, 75100, 24000,3500, 32000, 270000, 2300)
study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests", "Long_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOceanicBarrens","Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarrens")

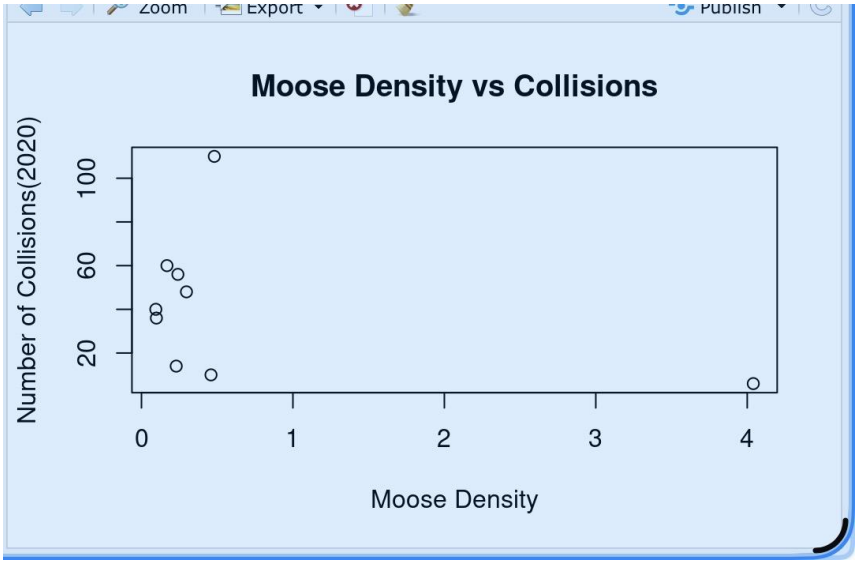
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
 >print(moose_coll)
Q25) > moose_coll2 <- moose_coll %>%
+ rename(Ecoregion = study_sites)
> coll_merge <- left_join(moose_coll2, moose_2020b, by = "Ecoregion", relationship = "many-to-many")
> print(coll_merge)
Q26a) > plot(coll_merge$MooseDensity,coll_merge$collisions2020,xlab = "Moose Density", ylab = "Number of Collisions(2020)", main = "Moose Density vs Collisions")
Q26b) There is a positive relationship between the number of collisions and the moose density. Higher moose density regions experience more collisions, while lower density regions experience less collisions. There can be an outlier where the collisions are very high even though there is average density.
Q27) > coll_merge_per_capita <- coll_merge %>%
+ mutate(coll_per_capita = collisions2020 / human_pop)
> print(coll_merge_per_capita)
Q28) > coll_merge_per_capita <- coll_merge %>%
+ mutate(coll_per_capita = collisions2020 / human_pop)
> print(coll_merge_per_capita)
Q29) Here it looks like the areas with a lower human population seem to have high collisions per person. It makes sense relating to Newfoundland as Newfoundland has a lower population and a surplus of moose. Every person is likely to encounter a moose in their lifetime a few times.

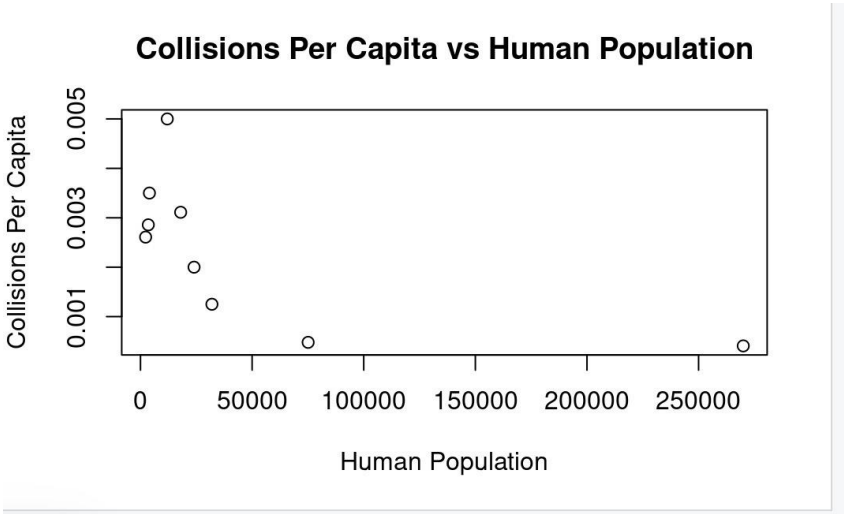
All graphs created are below:


Moose density in Newfoundland ecoregions over time
Moose per sq km
0.0
1.0
2.0
1900
1920
1940
1960
1980
2000
2020
year


Average Black Spruce Browsing by Ecoregion
Average Black Spruce Browsing Score
0
1
2
3
4
Avalon_Forests
Long_Range_Barrens
Northern_Peninsula_Forests
Ecoregion


Average_Balsam_Fir_Browsing_by_Ecoregion
Average_Balsam_Browsing_Score
0
1
2
3
4
Avalon_Forests
EasternHyperOceanicBarrens
Maritime_Barrens
Northern_Peninsula_Forests
Ecoregion


Zoom
Export
Publish
Moose Density vs Collisions
Number of Collisions(2020)
20
60
100
0
1
2
3
4
Moose Density


Collisions Per Capita vs Human Population
Collisions Per Capita
0.001
0.003
0.005
0
50000
100000
150000
200000
250000
Human Population